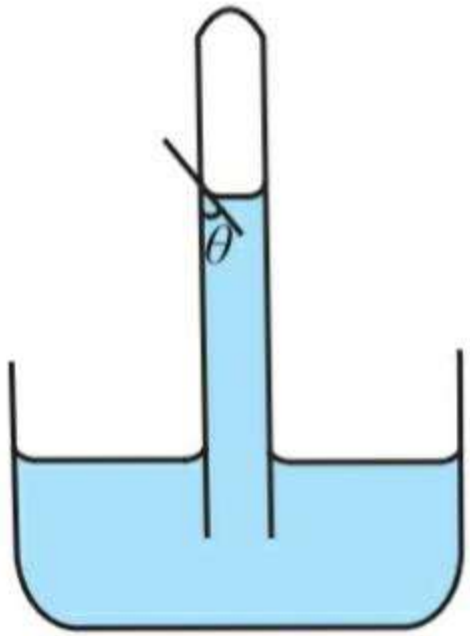
MECHANICAL PROPERTIES OF FLUID

PRESENTED
BY

WHAT IS ANGLE OF CONTACT ?

The angle of contact, θ , between a liquid and a solid surface is defined as the angle between the tangents drawn to the free surface of the liquid and surface of the solid at the point of contact, measured within the liquid.



ANGLE OF CONTACT DUE TO DIFFERENT SURFACES

CONCAVE MENISCUS

- When a liquid surface comes in contact with a solid surface, it forms a meniscus either concave or convex
- When the angle of contact is acute, the liquid forms a concave meniscus.

CONVEX MENISCUS

- When a liquid surface comes in contact with a solid surface, it forms a meniscus either concave or convex
- When the angle of contact is obtuse, it forms a convex meniscus

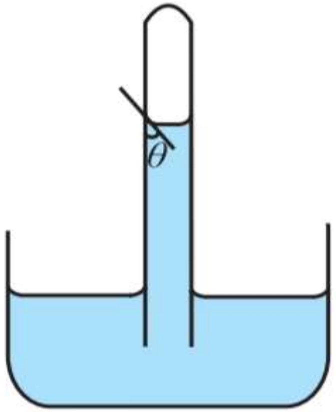


Fig. 2.18 (a): Concave meniscus due to liquids which partially wet a solid surface.

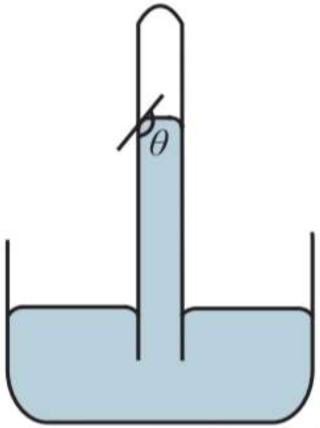


Fig. 2.18 (b): Convex meniscus due to liquids which do not wet a solid surface.

For example, water-glass interface forms a concave meniscus

For example,mercury-glass interface forms a convex meniscus.

DO YOU KNOW!!!

- When we observe the level of water in a capillary, we note down the level of the tangent to the meniscus inside the water.
- When we observe the level of mercury in a capillary we note down the level of the tangent to the meniscus above the mercury column

Angle of Contact

a) Shape of meniscus

b) Shape of liquid drops on a solid surface

A. Shape of meniscus

I) CONCAVE MENISCUS - ACUTE ANGLE OF CONTACT:

- Figure shows the acute angle of contact between a liquid surface (e.g., kerosene in a glass bottle).
- Considerr a molecule such as A on the surface of the liquid near the wall of the container.
- The molecule experiences both cohesive as well as adhesive forces. In this case, since the wall is vertical, the net adhesive force (AP) acting on the molecule A is horizontal.

i) Concave meniscus - acute angle of contact:

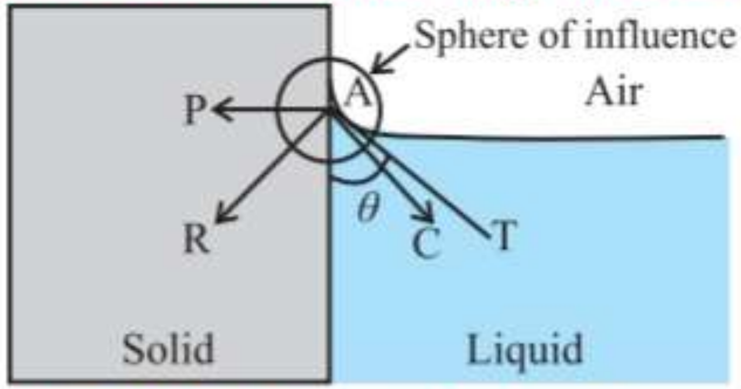


Fig. 2.19 (a): Acute angle of contact.

i) Concave meniscus - acute angle of contact:

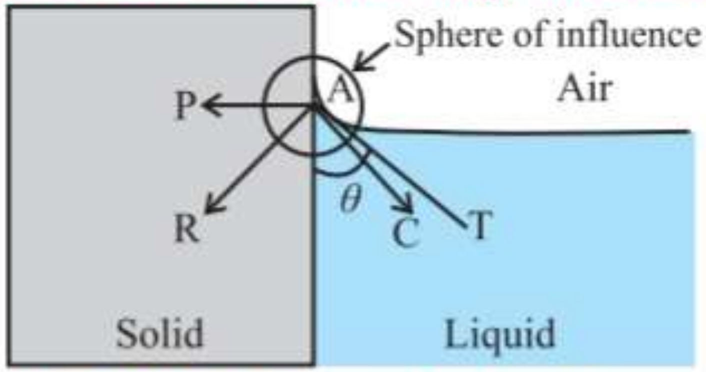


Fig. 2.19 (a): Acute angle of contact.

- Net cohesive force (AC) acting on molecule is directed at nearly 45° to either of the surfaces. Magnitude of adhesive force is so large that the net force (AR) is directed inside the solid.
- For equilibrium or stability of a liquid surface, the net force (AR) acting on the molecule A must be normal to the liquid surface at all points.
- For the resultant force AR to be normal to the tangent, the liquid near the wall should pile up against the solid boundary so that the tangent AT to the liquid surface is perpendicular to AR. Thus, this makes the meniscus concave.
- Obviously, such liquid wets that solid surface.

A. SHAPE OF MENISCUS

II) CONVEX MENISCUS - OBTUSE ANGLE OF CONTACT:

- Figure shows the obtuse angle of contact between a liquid and a solid (e.g., mercury in a glass bottle).
- Consider a molecule such as A on the surface of the liquid near the wall of the container.
- The molecule experiences both cohesive as well as adhesive forces. In this case also, the net adhesive force (AP) acting on the molecule A is horizontal since the wall is vertical.

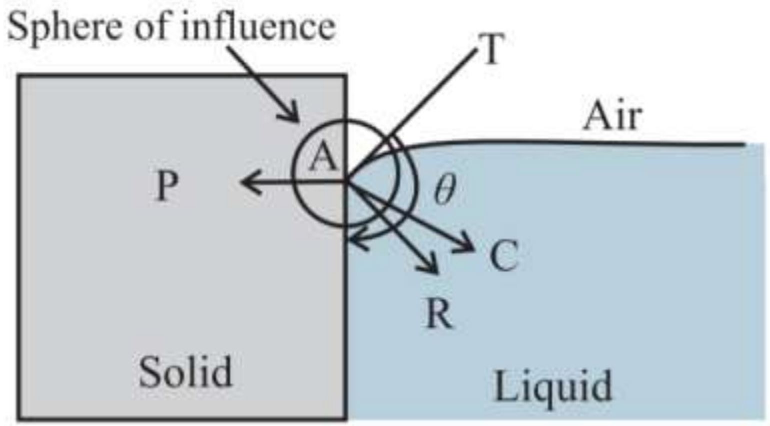


Fig. 2.19 (b): Obtuse angle of contact.

- Magnitude of cohesive force is so large that the net force (AR) is directed inside the liquid.

- For equilibrium or stability of a liquid surface, the net force (AR) acting on all molecules similar to molecule A must be normal to the liquid surface at all points.
- The liquid near the wall should, therefore, creep inside against the solid boundary. This makes the meniscus convex so that its tangent AT is normal to AR.
- Obviously, such liquid does not wet that solid surface.

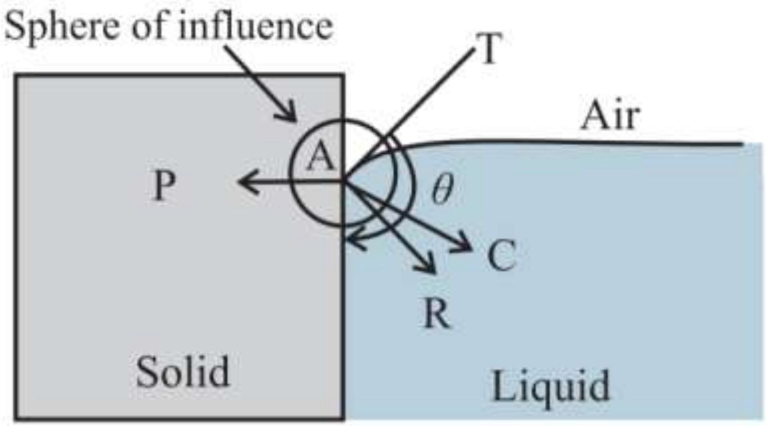


Fig. 2.19 (b): Obtuse angle of contact.

A. SHAPE OF MENISCUS

III) ZERO ANGLE OF CONTACT :

- Figure shows the angle of contact betwee a liquid (e.g. highly pure water) which completely wets a solid (e.g. clean glass) surface.
- The angle of contact in this case is almost zero (i.e., $\theta \to .\ 0^0$).
- In this case, the liquid molecules near the contact region, are so less in number that the cohesive force is negligible, i.e. AC =0 and the net adhesive force itself is the resultant force,i.e.AR = AP

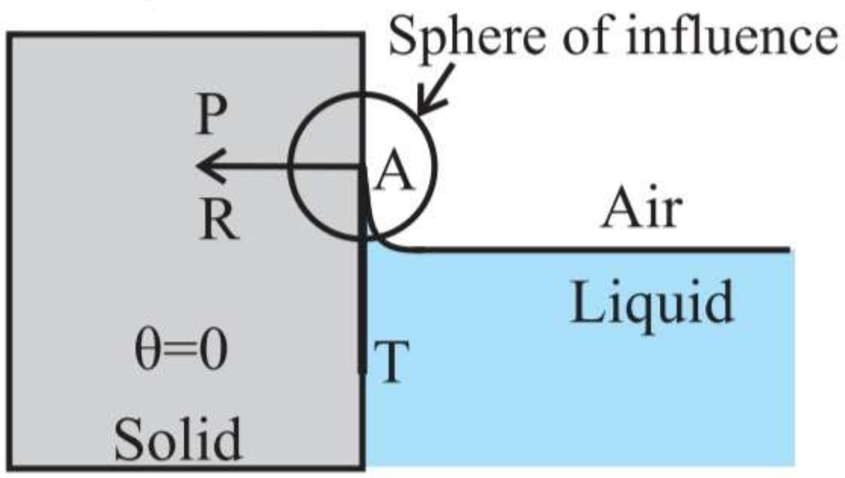


Fig. 2.19 (c): Angle of contact equal to zero.

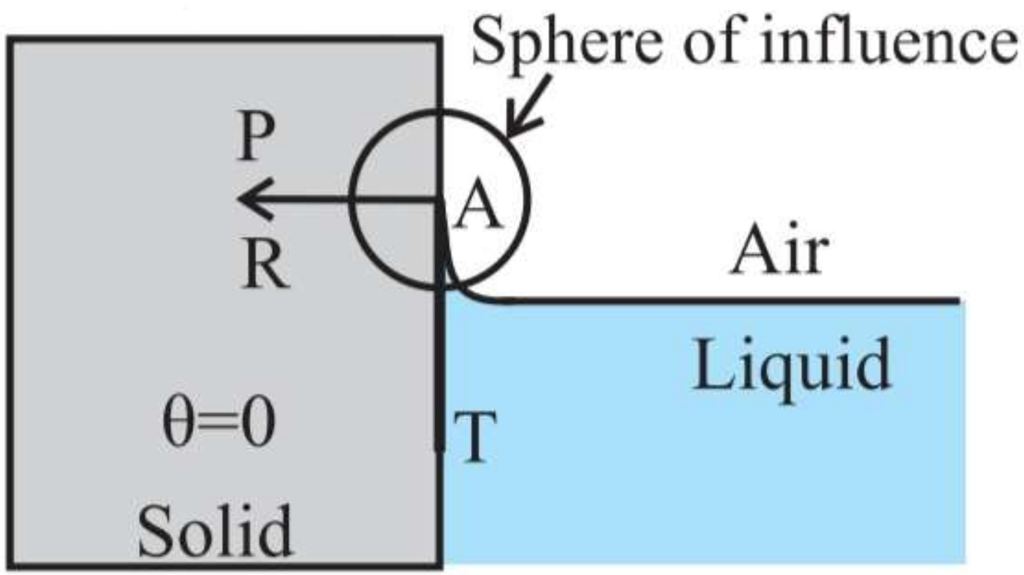


Fig. 2.19 (c): Angle of contact equal to zero.

Therefore, the tangent AT is along the wall within the liquid and the angle of contact is zero.

A. SHAPE OF MENISCUS

IV) Angle of contact 90^0 and conditions for convexity and concavity:

- Consider a hypothetical liquid having angle of contact 90^0 with a given solid container, as shown in the Fig. 2.19 (d).

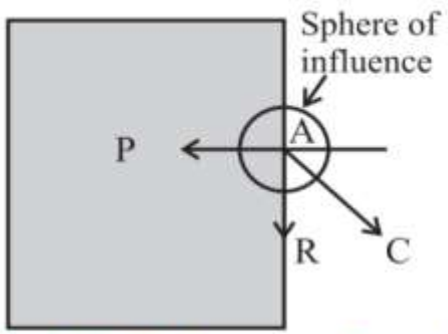


Fig. 2.19 (d): Acute angle equal to 90^0.

- In this case, the net cohesive force AC is exactly at 45^0 with either of the surfaces and the resultant force AR is exactly vertical (along the solid surface).

For this to occur, $\overrightarrow{AP} = \frac{AC}{\sqrt{2}}$ where, AR is the magnitude of the net force. From this we can write the conditions for acute and obtuse angles of contact:

For acute angle of contact, $\overrightarrow{AP} > \frac{AC}{\sqrt{2}}$, and for obtuse angle of contact, $\overrightarrow{AP} < \frac{AC}{\sqrt{2}}$.

Can You Tell ???

How does a water proofing agent work?

ANSWER

Water proofing changes the angle of contact

For example:If the clothes are sprayed with waterproofing agent, then it changes the angle of contact from acute to obtuse, in this way, water beads up on the surface and does not easily penetrate the clothing.

B) Shape of liquid drops on a solid surface:

- When a small amount of a liquid is dropped on a plane solid surface, the liquid will either spread on the surface or will form droplets on the surface.
- This phenomenon will occur depends on the surface tension of the liquid and the angle of contact between the liquid and the solid surface.

- Let θ be the angle of contact for the given solid-liquid pair.
- T1 = Force due to surface tension at the liquid- solid interface,
- T2 = Force due to surface tension at the air- solid interface,
- T3 = Force due to surface tension at the air- liquid interface.
- As the force due to surface tension is tangential to the surfaces in contact, directions of T1, T2 and T are as shown in the Fig.

- For equilibrium of the drop,

$$T_2 = T_1 + T_3 cos\theta \text{ , } cos\theta = \frac{T_2 - T_1}{T_3}$$

on the equation

$$T_2 = T_1 + T_3 cos\theta \ , cos\theta = \frac{T_2 - T_1}{T_3}$$

CASE 1

1) If $T_2 > T_1$ and $(T_2 - T_1) < T_3$, $\cos\theta$ is positive and the angle of contact θ is acute as shown in Fig. 2.20 (a).

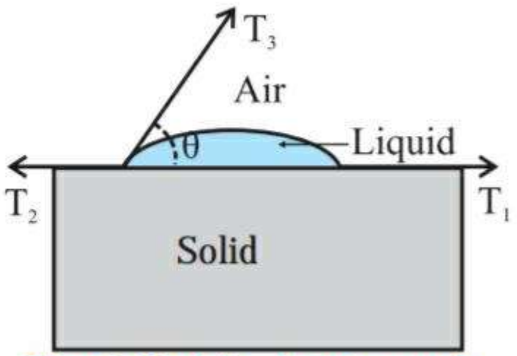


Fig. 2.20 (a): Acute angle.

CASE 2

2) If $T_2 < T_1$ and $(T_1 - T_2) < T_3$, $\cos\theta$ is negative, and the angle of contact θ is obtuse as shown in Fig. 2.20(b).

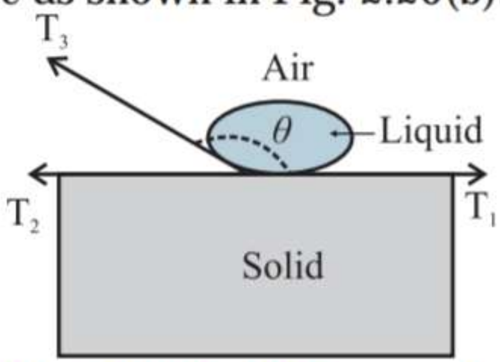


Fig. 2.20 (b): Obtuse angle.

on the equation

$$T_2 = T_1 + T_3 cos\theta \ , \ cos\theta = \frac{T_2 - T_1}{T_3}$$

CASE 3

3) If $(T_2 - T_1) = T_3$, $\cos\theta = 1$ and θ is nearly equal to zero.

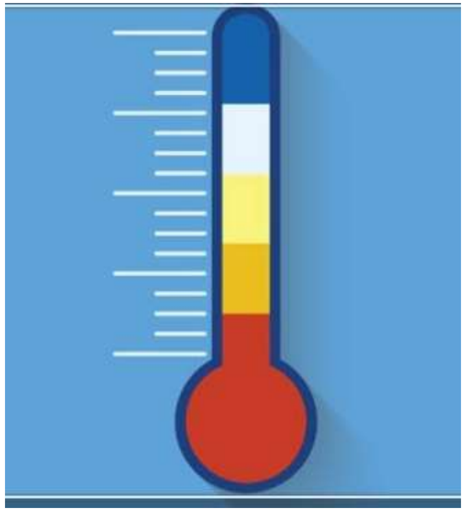
CASE 4

4) If $(T_2 - T_1) > T_3$ or $T_2 > (T_1 + T_3)$, $\cos\theta > 1$ which is impossible. The liquid spreads over the solid surface and drop will not be formed.

C) Factors affecting the angle of contact

- I) THE NATURE OF THE LIQUID AND THE SOLID IN CONTACT.

ii) Impurity : Impurities present in the liquid change the angle of contact.



- III) TEMPERATURE OF THE LIQUID : ANY INCREASE IN THE TEMPERATURE OF A LIQUID DECREASES ITS ANGLE OF CONTACT. FOR A GIVEN SOLID-LIQUID SURFACE, THE ANGLE OF CONTACT IS CONSTANT AT A GIVEN TEMPERATURE.

ANGLE OF CONTACT FOR PAIR OF LIQUID - SOLID IN CONTAC

Sr. No.	Liquid - solid in contact	Angle of contact
1	Pure water and clean glass	$0°$
2	Chloroform with clean glass	0^0
3	Organic liquids with clean glass	0^0
4	Ether with clean glass	16^0
5	Kerosene with clean glass	26^0
6	Water with paraffin	107^0
7	Mercury with clean glass	140^0

REVIEW

Shape of meniscus

- Concave meniscus - acute angle of contact
- Convex meniscus - obtuse angle of contact
- Zero angle of contact
- Angle of contact $90^0$and conditions for convexity and concavity:

Shape of liquid drops on a solid surface

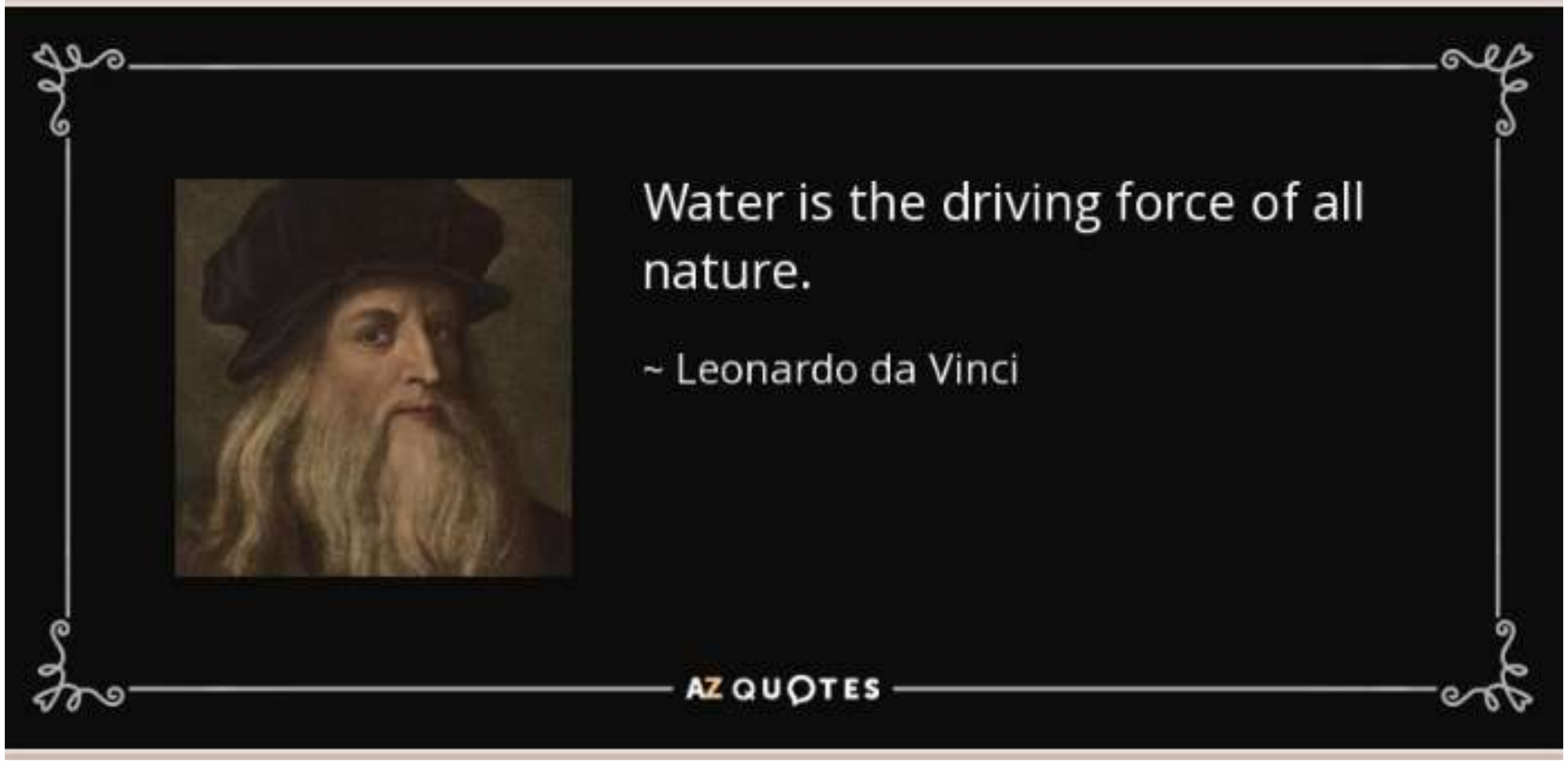
- When a small amount of a liquid is dropped on a plane solid surface, the liquid will either spread on the surface or will form droplets on the surface.
- This phenomenon will occur depends on the surface tension of the liquid and the angle of contact between the liquid and the solid surface.
- For equilibrium of the drop,

$$T_2 = T_1 + T_3 cos\theta \text{ , } cos\theta = \frac{T_2 - T_1}{T_3}$$

Factors affecting the angle of contact

- The nature of the liquid and the solid in contact.
- ii) Impurity : Impurities present in the liquid change the angle of contact.
- iii) Temperature of the liquid : Any increase in the temperature of a liquid decreases its angle of contact. For a given solid-liquid surface, the angle of contact is constant at a given

THANKS


Water is the driving force of all nature.
~ Leonardo da Vinci
AZ QUOTES